



Introduction to Statistics and Probability

Exercise sheet # 2

Exercise 1 – Let X be a binomial r.v. with parameter (n, p) . Compute the moment-generating function $M_X(t)$ and use it to find $E(X)$ and $Var(X)$.

Exercise 2 – Let $X \sim \mathcal{B}(m, p)$ $Y \sim \mathcal{B}(n, p)$ with X and Y independent r.v.'s and $m \geq n$.

1. Show that $Z = X + Y \sim \mathcal{B}(m + n, p)$.
2. Find the conditional distribution of X given $Z = X + Y$.

Exercise 3 – Let X and Y be two independent r.v. X has a Poisson distribution with parameter λ_1 and Y has a Poisson distribution with parameter λ_2 .

1. Show that $X + Y$ has a Poisson distribution with parameter $\lambda_1 + \lambda_2$.
2. Show that the conditional distribution of X given $X + Y$ is a binomial distribution.

Exercise 4 – Let us consider $X_1, X_2, \dots, X_n$, n v.a. independent identically distributed from a Poisson distribution with parameter λ .

Find the distribution of $Y = X_1 + X_2 + \dots + X_n$.

Exercise 5 – The number X of electrons emitted by a radioactive item in a given time window has a Poisson distribution with parameter λ :

$$P(X = n) = \frac{\lambda^n}{n!} e^{-\lambda}, \quad n \in \mathbb{N}, \quad \lambda > 0.$$

Each electron has a probability p for having a biological effect. Let us denote Z the number of electrons having a biological effect among the electrons emitted by the radioactive item.

1. What is the probability that among the emitted electrons, k have a biological effect that is to say : $P(Z = k | X = n)$?
2. Deduce from the question above the distribution of Z .

Exercise 6 – Let Y be a r.v. that has a Poisson distribution with a parameter $\lambda > 0$ and X a r.v. such that X given $\{Y = y\}$ has a binomial distribution with parameters (y, p) , $(0 < p < 1)$.

1. Show that the distribution of X is a Poisson distribution with parameter λp .
2. Show that the distribution of $Y - X$ is a Poisson distribution with parameter $\lambda(1 - p)$.
3. Are X and $Y - X$ independent?

Exercise 7 – Let X be the uniform r.v. defined on $(0, 1)$ that is with p.d.f. $f_X(x) = \mathbb{1}_{(0,1)}(x)$ where $\mathbb{1}_{(0,1)}(x) = 1$ if $x \in (0, 1)$ that is to say $0 < x < 1$ and 0 otherwise.

1. Find $F_X(x)$, the c.d.f. of X .
2. Find $f_Y(y)$, the p.d.f. of $Y = -2\log X$ (give the name and the parameter).

Exercise 8 – Consider X a r.v. having a Weibull distribution with parameters (α, β) . Its p.d.f. is :

$$f_X(x) = \frac{\beta}{\alpha^\beta} x^{\beta-1} \exp \left\{ - \left(\frac{x}{\alpha} \right)^\beta \right\}, \quad \alpha, \beta > 0.$$

1. Express the cumulative distribution function (c.d.f.).
2. Calculate the moment of order m .
3. Deduce the expression of the expectation and the variance.

Exercise 9 – Show that if a random variable X is exponential then it has the *lack of memory property*. That is to say

$$P(X > x_1 + x_2 \mid X > x_1) = P(X > x_2)$$

Would it be true for a Weibull distribution?

Exercise 10 – Let the 3-dimensional vector $\underline{X} = (X_1, X_2, X_3)$ have the p.d.f. :

$$f_{\underline{X}}(x_1, x_2, x_3) = \begin{cases} 6x_1 x_2^2 x_3 & \text{if } 0 \leq x_1 \leq 1, 0 \leq x_2 \leq 1, 0 \leq x_3 \leq \sqrt{2} \\ 0 & \text{otherwise.} \end{cases}$$

1. Show that $f_{\underline{X}}(x_1, x_2, x_3)$ is a pdf.

2. Find the marginal pdf's of X_1 , of X_2 , of X_3 , of (X_1, X_3) .
3. Find the conditional pdf's $f_{X_1, X_2 | X_3 = x_3}(x_1, x_2)$ and $f_{X_3 | X_1 = x_1}(x_3)$.
4. Are X_1 , X_2 and X_3 independent r.v.'s?

Exercise 11 – Let (X_1, X_2) be a two-dimensional r.v. with p.d.f. :

$$f_{\underline{X}}(x_1, x_2) = \begin{cases} x_1 + x_2 & \text{if } 0 \leq x_1 \leq 1, 0 \leq x_2 \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

1. Find the marginal pdf's of X_1 and X_2 .
2. Find the conditional pdf's X_2 given $X_1 = x_1$.
3. Are X_1 and X_2 independent r.v.'s?

Exercise 12 – Suppose that (X, Y) has the joint density function

$$f_{X,Y}(x, y) = \begin{cases} \frac{1}{4}(1 + xy(x^2 - y^2)), & \text{for } |x| < 1 \text{ and } |y| < 1 \\ 0, & \text{otherwise.} \end{cases}$$

Find the density of X ; of Y ; of $X + Y$.

Exercise 13 – If X and Y have the joint density function

$$f_{X,Y}(x, y) = \begin{cases} \frac{1}{8\pi^2} e^{-\frac{x^2+y}{4\pi}}, & \text{if } -\infty < x < +\infty, y \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

Find the density of $f_X(x)$ and $f_Y(y)$. Are X and Y independent?